Likelihood and Maximum Likelihood

Day 1: recap, the coin, the normal model, and where models come from

1. Recap

Recap Terminology

$f(x;\theta) =$ PDF of RV X with params θ

$\hat{\theta} =$ estimate of θ

$\bar{x} =$ sample mean $\rightarrow$ est. μ, $\qquad s^2 = \dfrac{\sum(x_i - \bar{x})^2}{n-1} \rightarrow$ est. σ^2

$$L = \prod_{i=1}^{n} f(y_i;\theta) = f(y_1;\theta) \times \cdots \times f(y_n;\theta) \Rightarrow \log L = \sum_{i=1}^{n} \log f(y_i;\theta)$$

$$\text{MSE} = \text{bias}^2 + \text{var} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Opening Question

- So far, estimates with something reasonable:
  - $\bar{x}$ for μ, s^2 for σ^2, proportion of sample → proportion in population
  - ECDF → CDF; KDE → PDF; transition matrix: count and divide
- But how wrong could it be? (uncertainty)
  - SE, CLT, bootstrap

Today: One General Recipe for Estimators

- (parametric, frequentist)

1. Pick $f(y;\theta)$ (modeling assumption)
2. Write the joint density over y and θ for the given data = **likelihood**
3. Choose θ to maximize ℓ

- Data X is an RV, not θ!
- Most methods are MLE for particular problems: OLS, logistic regression, k-means clustering, GMM, etc.

2. The coin

Likelihood as a function of the parameter

Maximum Likelihood Estimation

- MLE: pick params that are most likely to have resulted in our observed sample
- Assumptions: an **IID sample** and a **picked distribution family**
- WHY? Nice properties (consistent; even though it is not always unbiased)
- Essentially plug and play: one formula
- Works for parameter estimation, curve fitting, etc.
- Underlies major model-fitting packages (statsmodels, scikit-learn)

Likelihood: What Parameter Values Explain the Data?

$$L(p) \approx P(\text{data} \mid p)$$

- L is a **function of p**
- X is a random variable. The data is a sample of X

Weighted Coin Example (1 Binomial sample)

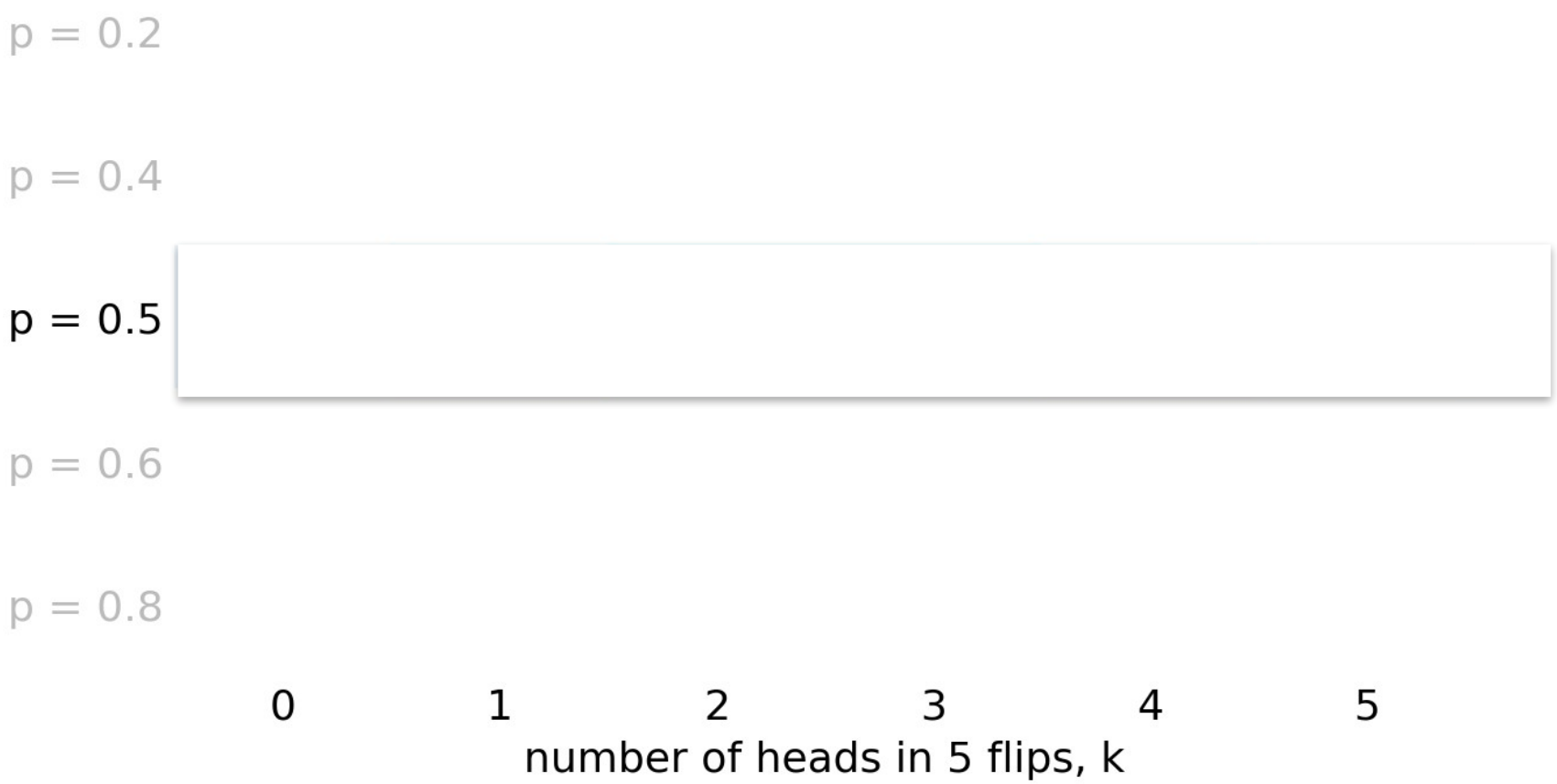
- **RV:** heads in 5 coin flips

$$P(X = k) = \binom{5}{k} p^k (1-p)^{5-k}$$

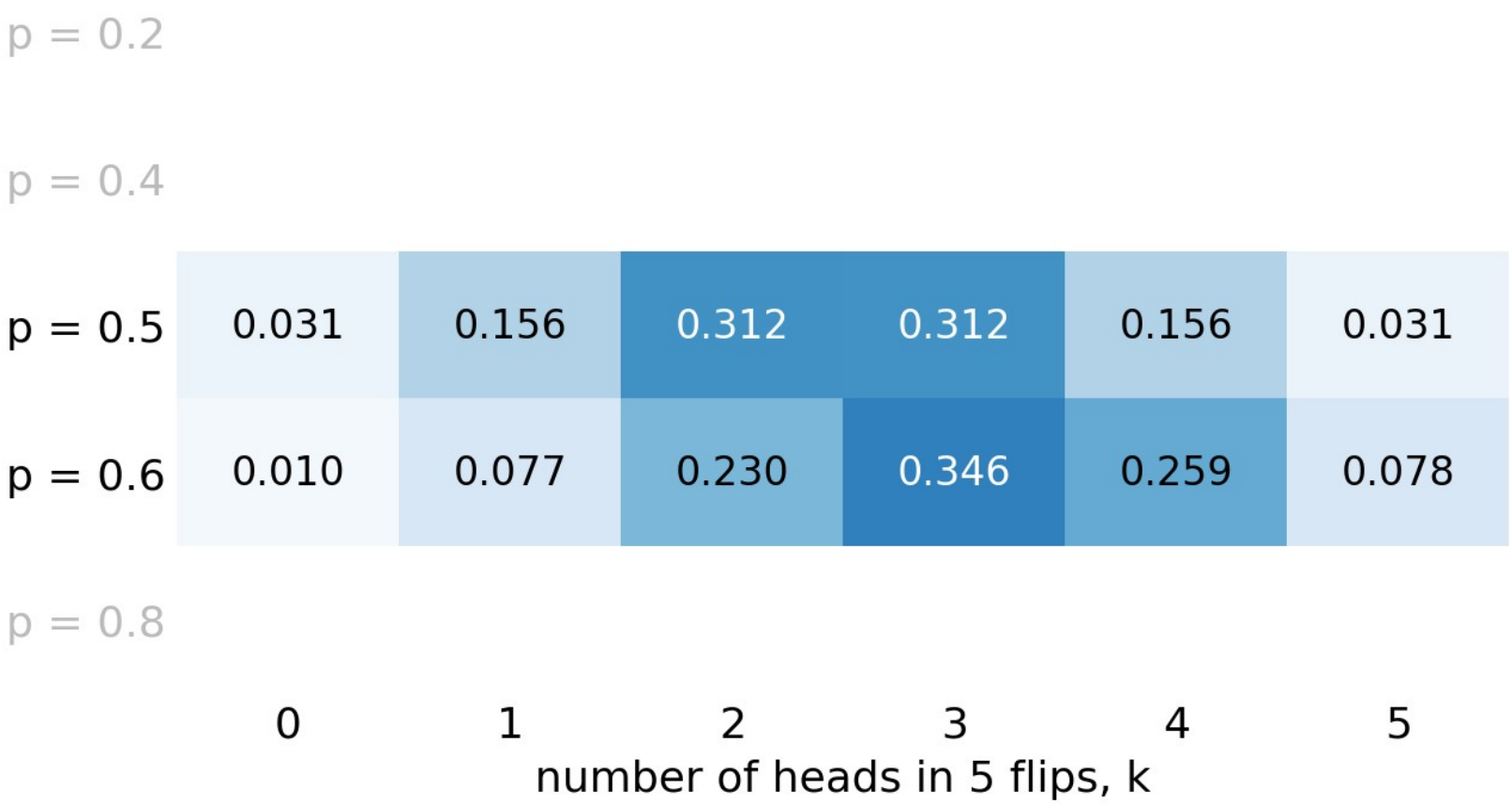
$$= \binom{5}{k} (0.5)^5 = \binom{5}{k} \frac{1}{32} \quad (p = 0.5)$$

Coin PMF Table: p = 0.5

- Start with a familiar PMF: p = 0.5. It sums to 1

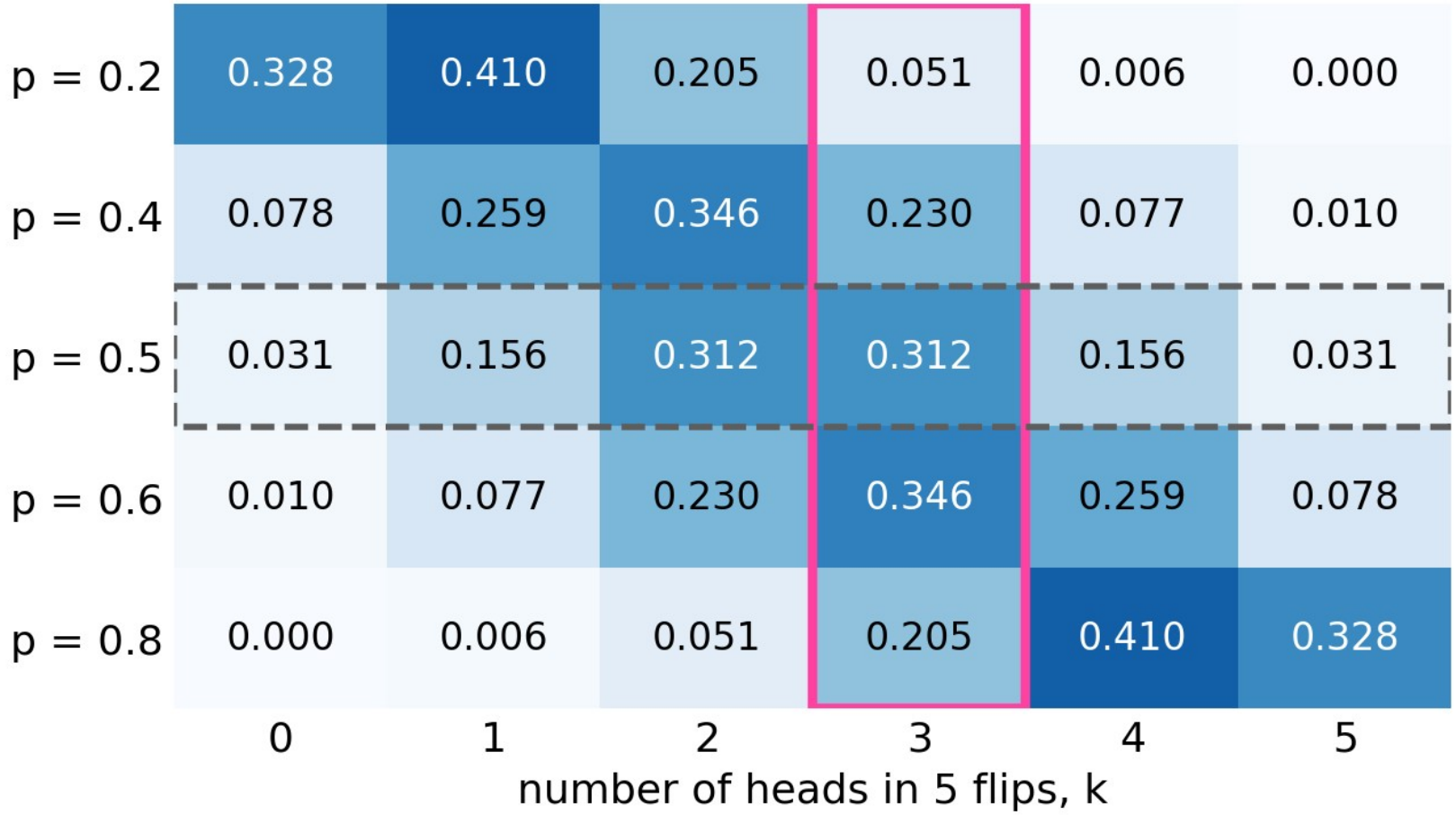


Coin PMF Table: Change p, Hold the Data Fixed



- Changing p, but the data we got stays the same:

Coin PMF Table: All Values of p



- Now consider the data: **we got 3/5 heads** (one observation)
- Look at the k = 3 column: Pr(3/5 | p = 0.5) = 0.312 and Pr(3/5 | p = 0.6) = 0.346
- Each row sums to 1. The pink column is the one we observed (k = 3)
- Down that column the largest value is at p = 0.6 (0.346), next to p = 0.4 (0.230) and p = 0.8 (0.205)

Likelihood Is the Column for the Data We Got

- Read down the k = 3 column over p: the max is at **p = 0.6**
- This is the **sample proportion**: MLE = 3/5
- Table = conditional density $f(x \mid \theta)$, one row per θ

- What about 3/5 heads, then 2/5 heads?
- Two observations → multiply the likelihoods

The Coin Table, Several Observations

	obs 1: 3/5 heads	obs 2: 2/5 heads	obs 3: 4/5 heads	joint = product = L(p)
p = 0.2	0.051	0.205	0.006	6.71e-05
p = 0.4	0.230	0.346	0.077	6.12e-03
p = 0.5	0.312	0.312	0.156	1.53e-02
p = 0.6	0.346	0.230	0.259	2.06e-02
p = 0.8	0.205	0.051	0.410	4.29e-03

Columns are observations (5 flips each). Each cell is the binomial pmf f(k | p) for that count

Each row's **product** is the joint density of the three observations, L(p). The pink column is the likelihood

The largest product is at p = 0.6 = 9/15 (all 15 flips pooled)

Maximum Likelihood

$\max_{\theta} L(\theta)$ is saying "pick parameters to make observations as likely as possible"

- L(θ) is the PDF of X viewed as a function of θ
- L(θ) is **NOT** a probability that θ is correct (θ is not random: no distribution, no probability)
- Only ratios of L are meaningful
- Probability over θ → Bayes (a different question; not today)
- Continuous data → use density, not probability.

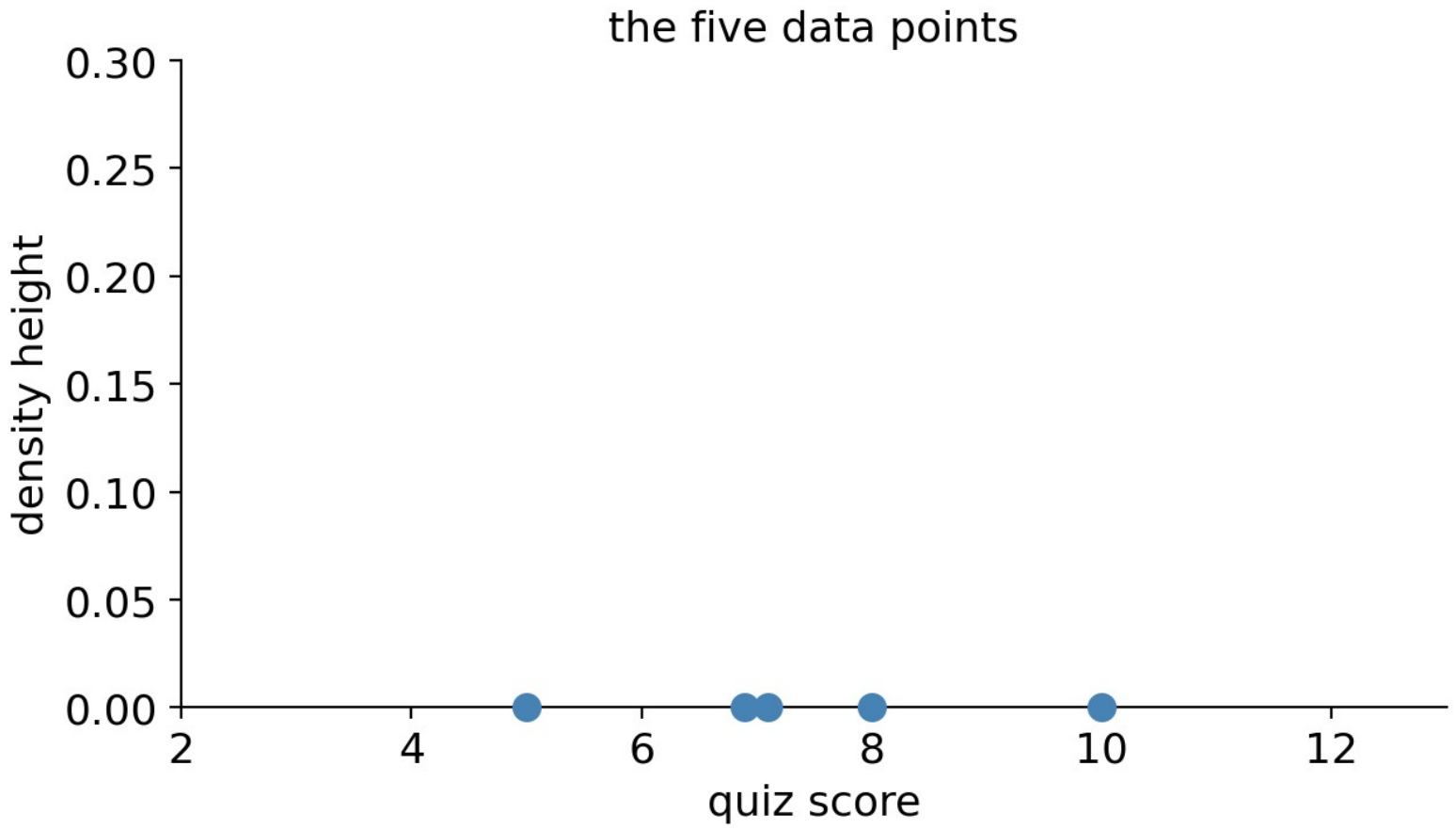
$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta) \quad \text{(the joint density of the data under independence)}$$

How to solve an ML estimation problem

- Write down the likelihood $\mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x})$.
- Maximize the likelihood by solving $\widehat{\boldsymbol{\theta}}_{\mathrm{ML}} = \operatorname*{argmax}_{\boldsymbol{\theta}} \log \mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x})$.

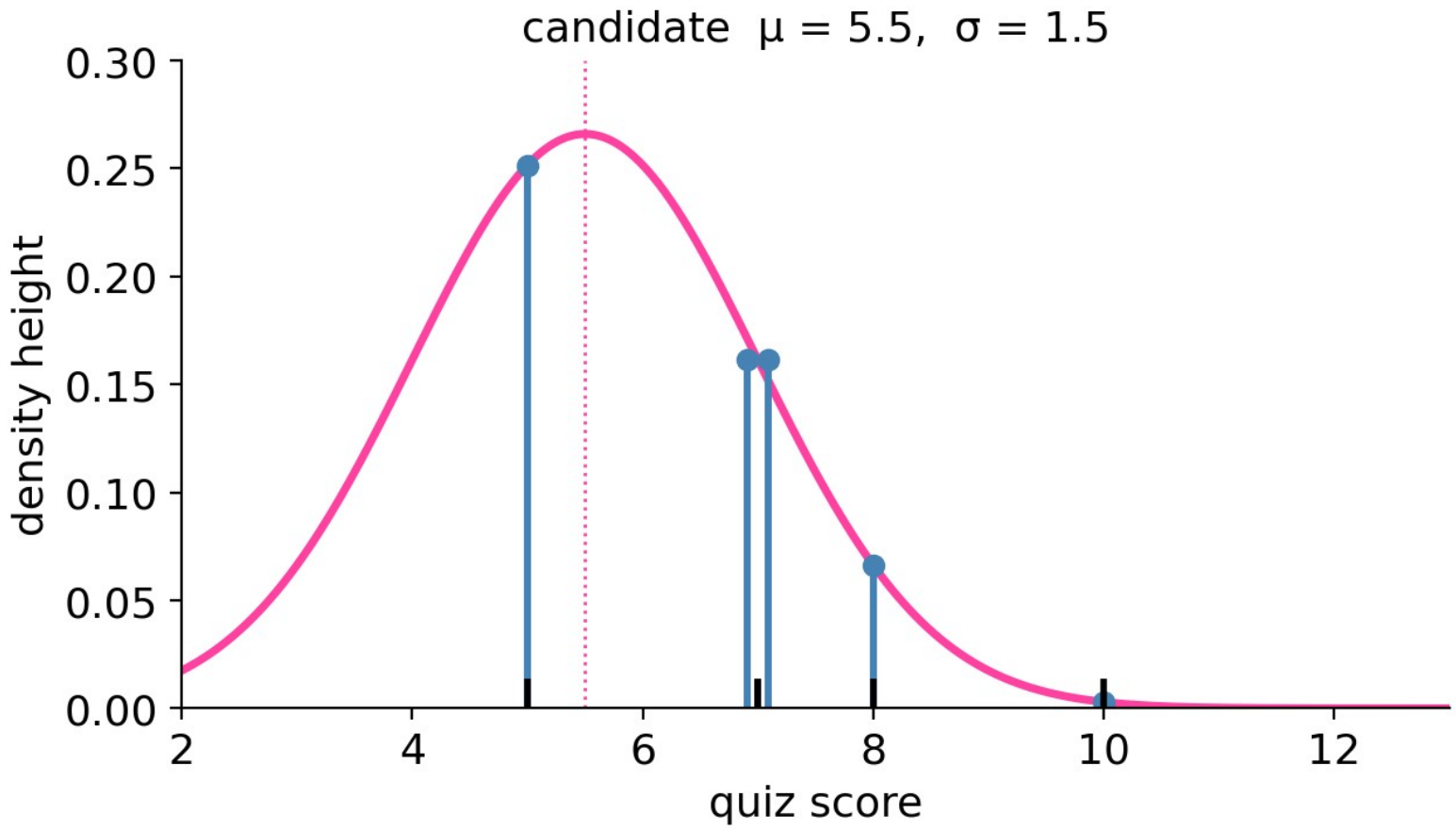
3. MLE for normally distributed variables

Five Data Points



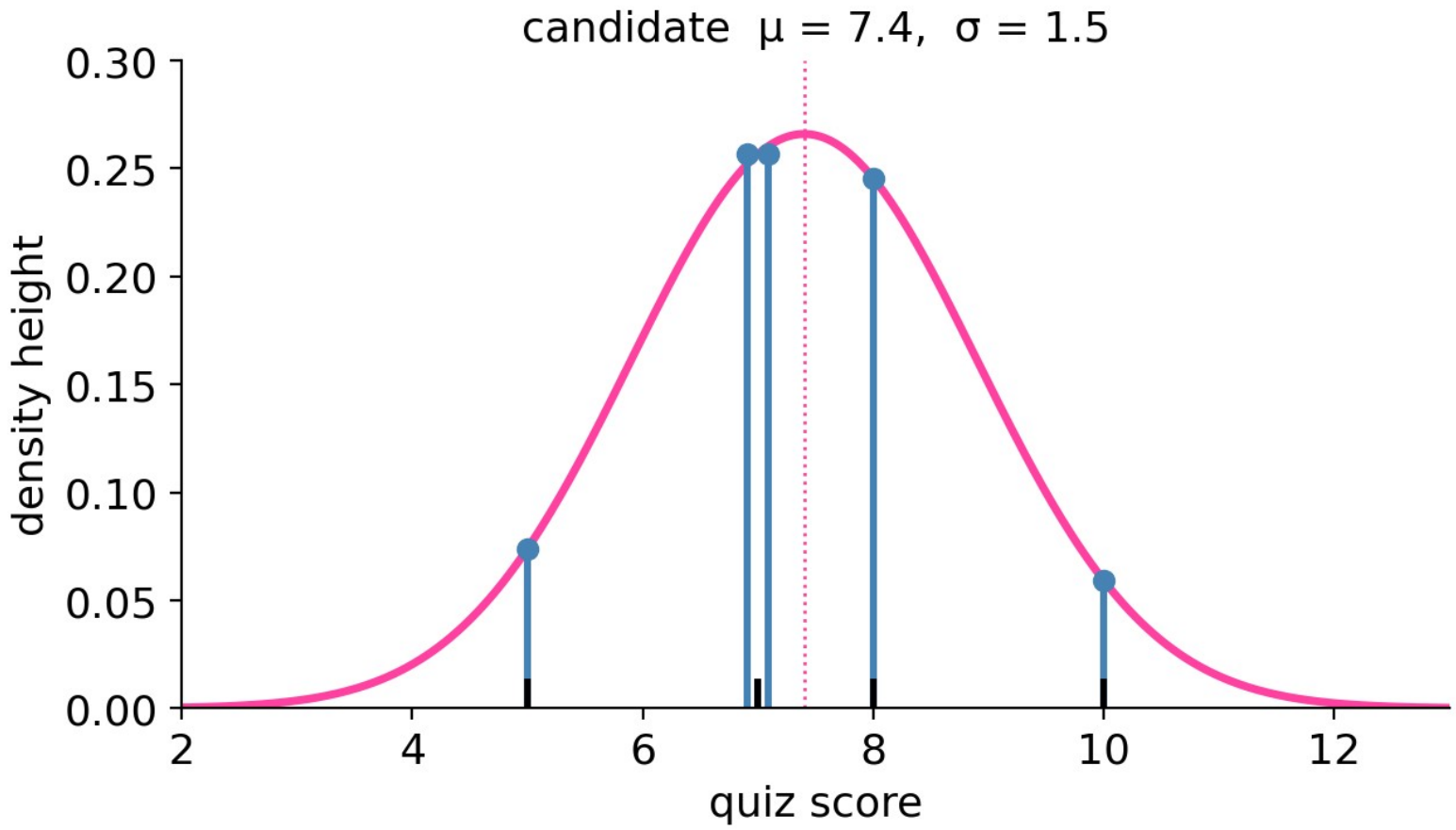
- Five quiz scores: 5, 7, 7, 8, 10
- Next: put a normal over each one, and slide μ
- Assume fixed variance, just finding estimate of μ

Sliding Normal: μ = 5.5



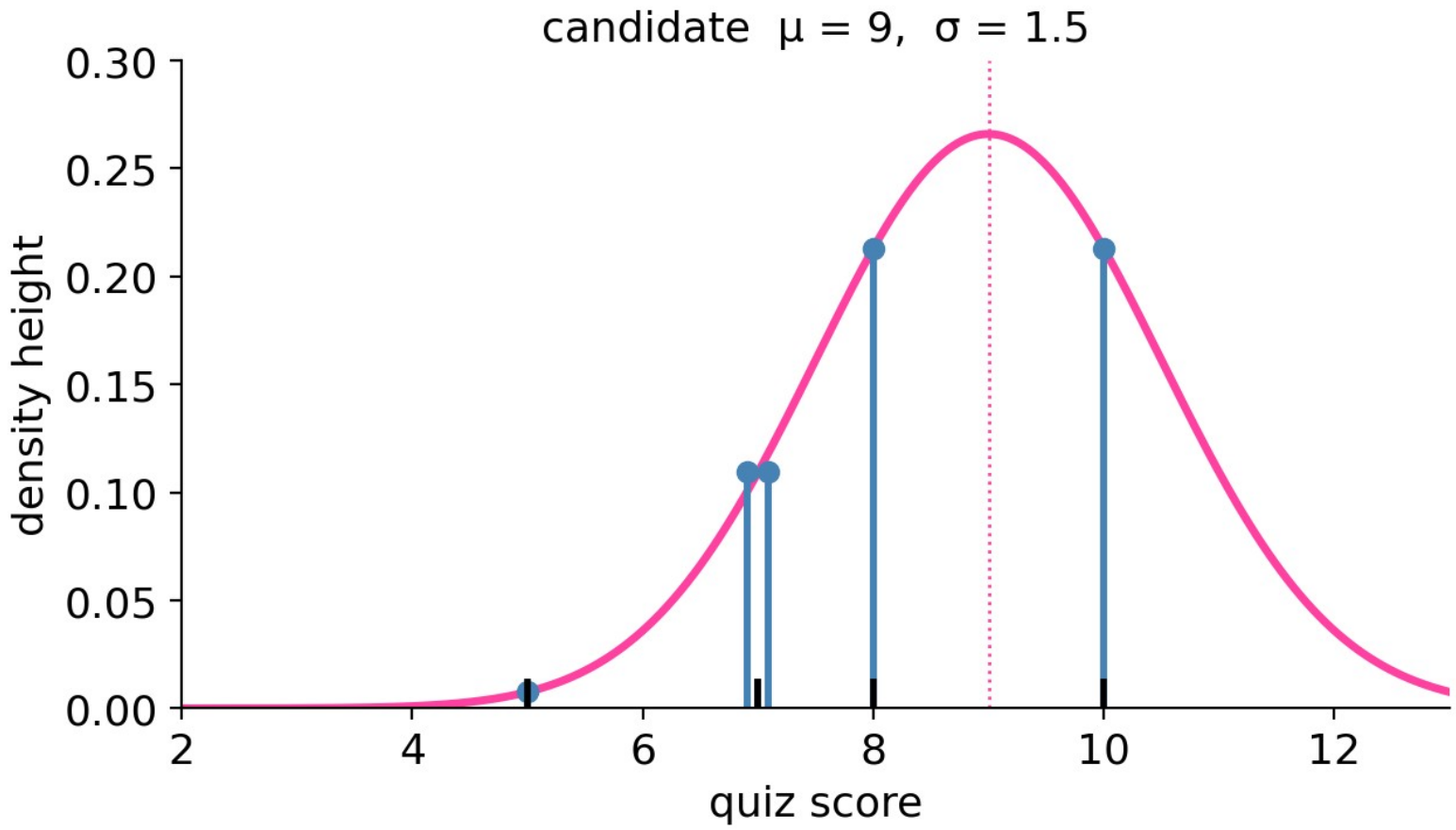
- Normal intuition first
- $f(x_i \mid \mu, \sigma) = 1 / (\sigma\sqrt{(2\pi)}) \cdot \exp(-(x_i - \mu)^2 / (2\sigma^2))$
- Slide μ to 5.5 and watch the density move over the data

Sliding Normal: μ = 7.4



- Normal intuition first
- $f(x_i \mid \mu, \sigma) = 1 / (\sigma\sqrt{(2\pi)}) \cdot \exp(-(x_i - \mu)^2 / (2\sigma^2))$
- Slide μ to 7.4 and watch the density move over the data

Sliding Normal: μ = 9



- Normal intuition first
- $f(x_i \mid \mu, \sigma) = 1 / (\sigma\sqrt{(2\pi)}) \cdot \exp(-(x_i - \mu)^2 / (2\sigma^2))$
- Slide μ to 9 and watch the density move over the data

The PDF table at specific values, for the Normal

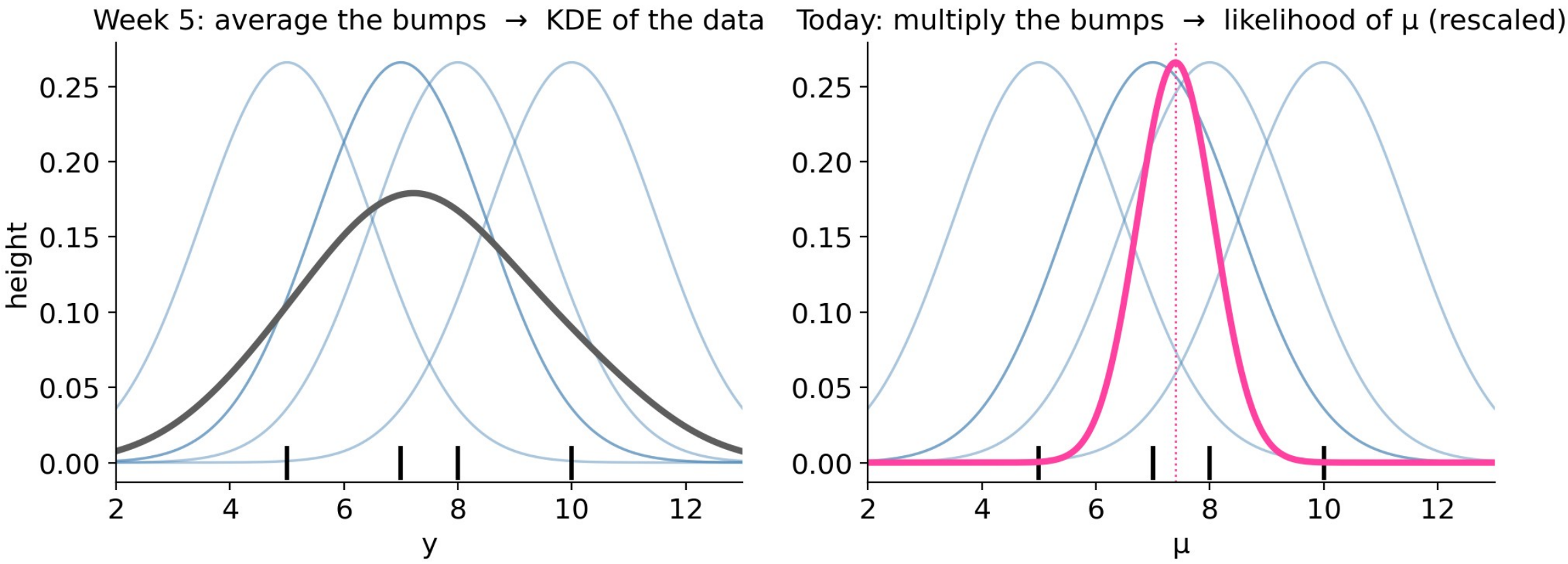
	x = 5	x = 7	x = 7	x = 8	x = 10	product = L(μ)
μ = 5.5	0.252	0.161	0.161	0.066	0.003	1.3e-06
μ = 7.4	0.074	0.257	0.257	0.246	0.059	7.1e-05
μ = 9	0.008	0.109	0.109	0.213	0.213	4.1e-06

Rows: candidate μ. Columns: the five data points. Each cell is the density f(x_i | μ), the stem height in the sliding frames

A row's **product** is the joint density of the sample for that μ, so the last column is L(μ) = ∏ f(x_i | μ)

Pink column: the likelihood. Read down it to see which μ makes the data most likely (the sliding normal peaks near μ = 7.4)

Normal Over Each Data Point



Step 1: Likelihood of the Normal Sample

- 1. Density of one observation

$$f(x_i \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

- 2. Likelihood: the product over the sample

$$L(\mu, \sigma) = \prod_{i=1}^{n} f(x_i \mid \mu, \sigma)$$

- We are planning to maximize → take logs to make it easier

Taking Logs: the Product Becomes a Sum

- log(ab) = log a + log b log(a^k) = k log a log(e^{a_i}) = a_i

$$L(\mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

$$\ell(\mu, \sigma) = -n\log\sigma - \frac{n}{2}\log(2\pi) - \frac{1}{2\sigma^2}\underbrace{\sum_{i=1}^{n}(x_i - \mu)^2}_{\text{SSE}}$$

- **So, the log-likelihood** is the expression above. The SSE is the only term that depends on the data

Step 2: Maximize with Respect to Each θ

- Maximize L(θ) with respect to each θ
- Take the derivative and set it equal to 0, then solve for θ

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu)$$

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^{n} (x_i - \mu)^2$$

- Set each equal to zero. Start with μ, which does not need σ

Solving for μ̂

$\frac{\partial \ell}{\partial \mu} = 0$ and solve

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu)$$

$$\sum_{i=1}^{n} x_i - n\hat{\mu} = 0$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i = \bar{x} \quad \text{(sample mean)}$$

- Max likelihood = min sum of squared deviations

Solving for $\hat{\sigma}$

$$\frac{\partial \ell}{\partial \sigma} = 0 \quad \text{(with } \mu \text{ replaced by } \hat{\mu} = \bar{x}\text{)}$$

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^{n} (x_i - \mu)^2$$

Solving for $\hat{\sigma}$

$$\frac{\partial \ell}{\partial \sigma} = 0 \quad \text{(with } \mu \text{ replaced by } \hat{\mu} = \bar{x})$$

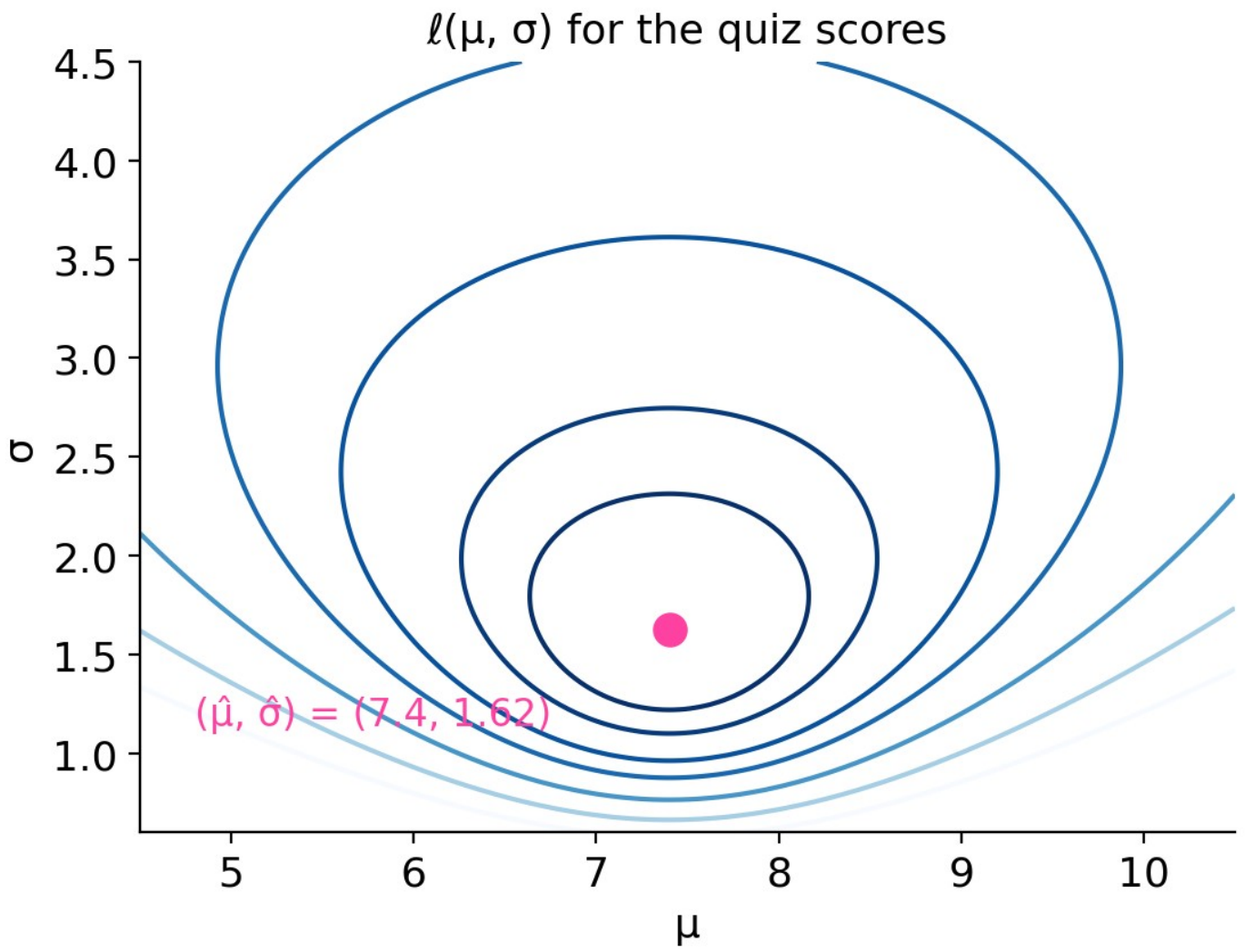
- Multiply by $\hat{\sigma}^3$:

$$n\hat{\sigma}^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

- **Sample variance!** The MLE $\hat{\sigma}^2$ is biased. s^2 has Bessel's correction, this does not

Both Parameters at Once



4. Where do models come from?

The mean-error decomposition

Where Do Models Come From?

- $X = \mu + \sigma\varepsilon$
  - model the error term explicitly (parametric)
  - link the model to data via parameters
  - MLE for estimate of parameters

Mean-Error Decomposition

- RV Y with mean μ and SE σ: y_i = μ + σε_i
- ε_i = shock, with mean 0, variance 1, density f(e)

$$\varepsilon_i = \frac{y_i - \mu}{\sigma}$$

- Rearrange and use the normal density

$$f(x_i \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$\log\left(\prod_i f(\varepsilon_i)\right) = \sum_i \log\left(f\left(\frac{y_i - \mu}{\sigma}\right)\right)$$

$$= -n\log\sigma - \frac{n}{2}\log(2\pi) - \frac{1}{2}\sum_i e_i^2$$

- For a fixed σ, maximizing L over μ ≡ **least squares**

Regression Is the Same Idea

Similarly, in regression: $y_i = x_i^\top \beta + \sigma \varepsilon_i$

Model μ by a linear function: replace $\mu_i \to x_i^\top \beta$

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Similarly, in regression: $y_i = x_i^\top \beta + \sigma \varepsilon_i$

Model μ by a linear function: replace $\mu_i \to x_i^\top \beta$

- Max over β ≡ least squares ≡ MLE under normal errors
- OLS = assumption of normality

$$L(\beta, \sigma) = -n \log\left(\sqrt{2\pi}\right) - n \log(\sigma) - \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^{n} (y_i - x_i\beta)^2}_{\text{Sum of Squared Error}}$$

(from normal)

$$\ell(\mu, \sigma) = -n \log \sigma - \frac{n}{2} \log(2\pi) - \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^{n} (x_i - \mu)^2}_{\text{SSE}}$$

Conditional Expectation

$$E[y_i \mid x_i] = E[x_i^\top \beta + \sigma\varepsilon \mid x_i] = x_i^\top \beta$$

- $x_i^\top \beta$ is the **conditional expectation function**!

Solving This MLE: Gradient Ascent

The gradient with respect to β is

$$\nabla_\beta L(\beta, \sigma) = \frac{1}{\sigma^2} X^\top (y - X\beta)$$

and with respect to σ^2 is

$$\frac{\partial L(\beta, \sigma)}{\partial \sigma} = \frac{-n}{\sigma} + \frac{(y - X \cdot \beta) \cdot (y - X \cdot \beta)}{\sigma^3}$$

- You can solve numerically by gradient ascent with

$$\begin{bmatrix} \beta_{t+1} \\ \sigma_{t+1} \end{bmatrix} = \begin{bmatrix} \beta_t \\ \sigma_t \end{bmatrix} + r \begin{bmatrix} \frac{1}{\sigma_t^2} X^\top (y - X\beta_t) \\ \frac{-n}{\sigma_t} - \frac{(y - X \cdot \beta_t) \cdot (y - X \cdot \beta_t)}{\sigma_t^3} \end{bmatrix}$$

- With the gradient → **solve numerically**

Aside: A Different Error

- What about Laplace distributed error?

Laplace density: $f(y \mid \mu, \gamma) = \frac{1}{2\gamma} \exp\left(-\frac{|y-\mu|}{\gamma}\right)$

- Fill in → L = sum of absolute deviations

$\hat{\mu} = \text{argmin} \sum |\text{dev}| = \text{median}$

- Median regression

Important: Loss Functions Are Densities

- Each loss function in regression is connected probabilistically to a **density**
- You cannot separate these conceptually

5. Confidence vs prediction intervals

CI vs PI

$$X_i = \mu + \sigma\varepsilon_i$$

- CI for μ:

$$\bar{x} \pm z\frac{\sigma}{\sqrt{n}}$$

- About the parameter μ. 95% of intervals contain μ (not a 95% chance of being in this interval)

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PI for a new observation: $X_{\text{new}} = \mu + \sigma\varepsilon_{\text{new}}$, independent of the data

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PI for a new observation: $X_{\text{new}} = \mu + \sigma\varepsilon_{\text{new}}$, independent of the data

$$X_{\text{new}} - \bar{x} = \sigma\varepsilon_{\text{new}} - \sigma\bar{\varepsilon}, \quad \text{Var}(X_{\text{new}} - \bar{x}) = \sigma^2 + \frac{\sigma^2}{n}$$

Large-Sample Approximation

$$\bar{x} \pm z\,\sigma\sqrt{1+\frac{1}{n}}$$

- Large sample approx → cross out $\sqrt{(1 + 1/n)}$
- As n → ∞, the CI shrinks; the PI does not
- PI covers new observation's noise and uncertainty in $\bar{x}$ → always bigger
- CI says where the mean is; PI says where the data are

Example: Ames House Prices

- Model: log(price) ~ log(area) + overall quality + full baths, fit by OLS on 80% of the Ames data (the rest is test data). OLS with normal errors is the MLE from earlier
- The fitted line predicts the log price. $R^2 = 0.767$ on the training data
- A **90% prediction interval** for one house is its predicted log price ± 1.645 $\hat{\sigma}$. It covers where a new house's price should fall
- It is not a confidence interval for the mean price, so it does not shrink as n grows: the house-to-house noise σ is always there
- Check: on the 20% test set, **92.3%** of the 90% intervals contain the true log price, close to the 90% target

Ames: The Fitted Model with CI in statsmodels

```
import statsmodels.formula.api as smf

train, test = train_test_split(df, test_size=0.2, random_state=42)
# C() creates category indicators, with the lowest level as the reference.
# The log-area coefficient is the price elasticity, holding these categories fixed.
model = smf.ols(
    'np.log(price) ~ np.log(area) + C(Q("Overall.Qual")) + C(Q("Full.Bath"))',
    data=train,
).fit()
model.summary()
```

OLS Regression Results

Dep. Variable:	np.log(price)	R-squared:	0.767
Model:	OLS	Adj. R-squared:	0.765
Method:	Least Squares	F-statistic:	546.1
Date:	Tue, 10 Nov 2026	Prob (F-statistic):	0.00
Time:	01:33:52	Log-Likelihood:	519.94
No. Observations:	2344	AIC:	-1010.
Df Residuals:	2329	BIC:	-923.5
Df Model:	14		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	8.1191	0.161	50.465	0.000	7.804	8.435
C(Q("Overall.Qual"))[T.2]	0.2218	0.115	1.921	0.055	-0.005	0.448
C(Q("Overall.Qual"))[T.3]	0.6320	0.105	6.032	0.000	0.427	0.837
C(Q("Overall.Qual"))[T.4]	0.8305	0.100	8.334	0.000	0.635	1.026
C(Q("Overall.Qual"))[T.5]	1.0497	0.099	10.604	0.000	0.856	1.244
C(Q("Overall.Qual"))[T.6]	1.1540	0.099	11.633	0.000	0.959	1.349
C(Q("Overall.Qual"))[T.7]	1.3127	0.100	13.191	0.000	1.118	1.508
C(Q("Overall.Qual"))[T.8]	1.5317	0.100	15.310	0.000	1.335	1.728
C(Q("Overall.Qual"))[T.9]	1.7877	0.102	17.573	0.000	1.588	1.987
C(Q("Overall.Qual"))[T.10]	1.8210	0.109	16.762	0.000	1.608	2.034
C(Q("Full.Bath"))[T.1]	-0.1756	0.063	-2.790	0.005	-0.299	-0.052
C(Q("Full.Bath"))[T.2]	-0.1236	0.063	-1.961	0.050	-0.247	1.47e-05
C(Q("Full.Bath"))[T.3]	-0.0010	0.069	-0.015	0.988	-0.137	0.135
C(Q("Full.Bath"))[T.4]	-0.1112	0.129	-0.860	0.390	-0.365	0.142
np.log(area)	0.3925	0.018	22.046	0.000	0.358	0.427

Omnibus:	511.772	Durbin-Watson:	2.042

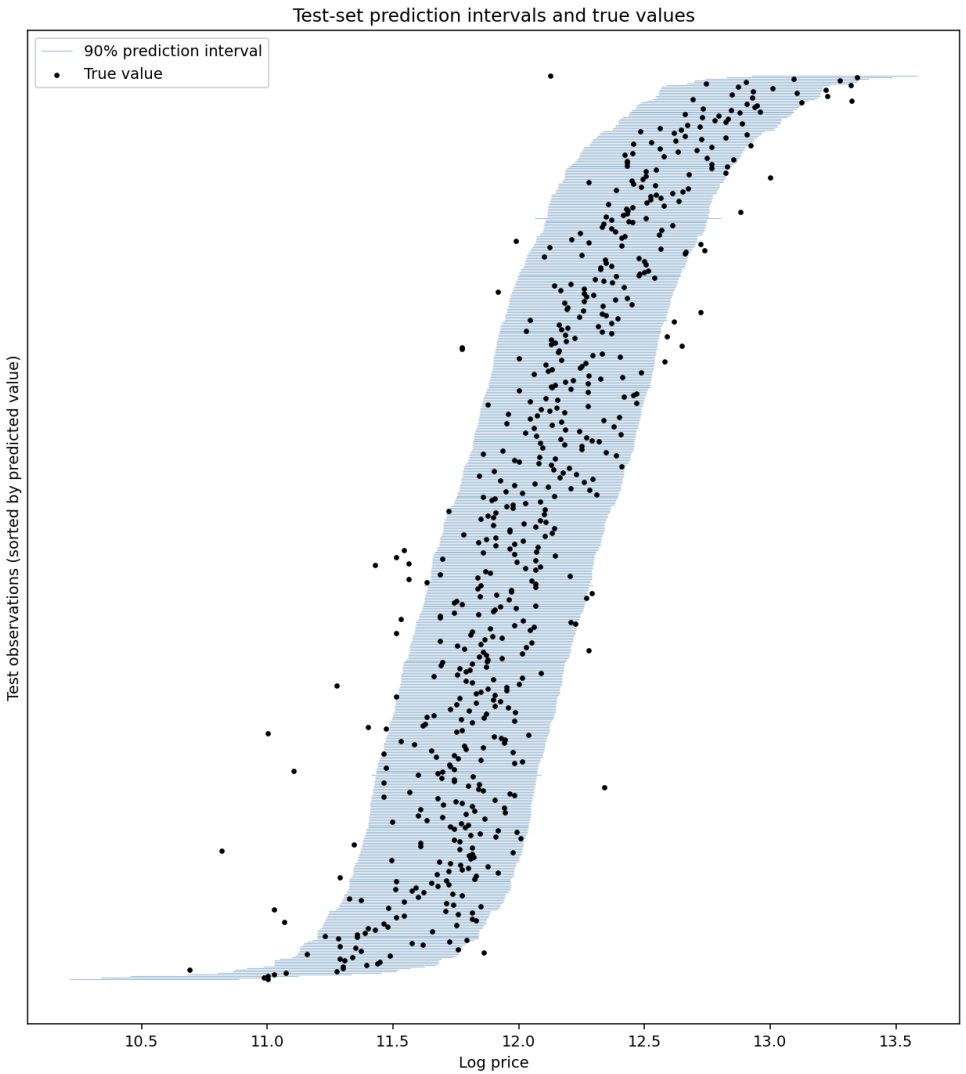
Ames: 90% Prediction Intervals in statsmodels

```
# 90% prediction intervals for individual log prices
prediction = model.get_prediction(test).summary_frame(alpha=0.10)
comparison = pd.DataFrame({
    'actual': np.log(test['price']),
    'predicted': prediction['mean'].to_numpy(),
    'lower': prediction['obs_ci_lower'].to_numpy(),
    'upper': prediction['obs_ci_upper'].to_numpy(),
})
coverage = comparison['actual'].between(comparison['lower'], comparison['upper']).mean()
print(f'Test-set coverage of 90% prediction intervals: {coverage:.1%}')
```

Output: Test-set coverage of 90% prediction intervals: 92.3%

	actual	predicted	lower	upper
1357	11.989160	12.386541	12.065573	12.707509
2367	11.661345	11.820141	11.499755	12.140526
2822	12.188418	12.232951	11.912615	12.553288
2126	11.724806	11.481063	11.160159	11.801967
1544	11.744037	11.775390	11.454938	12.095841

Ames: 90% Prediction Intervals on the Test Set



Each row is a test house, sorted by predicted value. Blue bar = 90% prediction interval; black dot = true log price. Most dots land inside their bars.